WORKSHEET

Chapter 11: Calculating and interpreting summary statistics

Questions

1 The following data was collected for the heights (in cm) of Year 8 students:

154 151 163 159 160 191 165 153 142 151 163

Answer these questions to one decimal place:

a Calculate and interpret the median score?

b Calculate and interpret the interquartile range?

c Calculate and interpret the mean.

d Calculate the standard deviation.

e Find any outliers.

f Recalculate the mean without the outlier(s).

g Recalculate the standard deviation without the outlier(s).

h Recalculate the interquartile range without the outlier(s).

i What is the difference between the means calculated with and without the outliers?

© Cengage Learning Australia Pty Ltd 2013

j What is the difference between the standard deviations calculated with and without the outliers?

k What is the difference between the interquartile ranges calculated with and without the outliers?

l Does the outlier(s) have more effect on the interquartile range or the standard deviation?

2 Some pumpkins are weighed and the weights (in kg) are recorded below:

2.3 2.7 2.4 3.1 0.5 2.8 2.4 3.3 3.2 2.7

Answer these questions to two decimal places:

a Calculate and interpret the mean of the data.

b Calculate and interpret the median.

c Calculate the standard deviation.

d Find any outliers.

e Recalculate the mean without the outlier(s).

f Recalculate the standard deviation without the outlier(s).

g What is the difference between the means calculated with and without the outlier(s)?

h Is the mean or median a better measure of centre for this set of scores?

i What effect does the outlier have on the standard deviation?

j Suggest a better measure of spread than the standard deviation for this data set.

© Cengage Learning Australia Pty Ltd 2013

3 The average hourly wage (in $) earned by a group of 18-year-olds is recorded below:

9 11 10 14 16 12 14 20 39 17 15 18 14

Answer these questions to one decimal place:

a Calculate and interpret the mean.

b Calculate and interpret the standard deviation.

c Find the median.

d Find any outliers.

e Recalculate the mean without the outliers.

f Recalculate the standard deviation without the outliers.

g Which is a more appropriate measure of centre for this data set – the mean or the median? Why?

h What effect does removing the outlier have on the standard deviation of these scores?

© Cengage Learning Australia Pty Ltd 2013

Answers

1 a The middle height of the students is 159 cm, which indicates that 50% of students are shorter than 159 cm and 50% are taller than 159 cm.
b The heights of the students have an IQR of 12 cm, which tells us that the middle 50% of heights has a range of 12 cm.
c The average height of the Year 8 class is 159.3 cm.
d 12.5 cm
e 191 cm ($133 \leq x \leq 181$)
f 156.1 cm
g 7.2 cm
h 12 cm
i The mean is 2.9 cm lower when the outlier is removed
j The standard deviation is 5.3 less when the outlier is removed
k The interquartile range does not change when the outlier is removed
l standard deviation

2 a The average weight of the pumpkins is 2.54 kg.
b The middle weight of the pumpkins is 2.7 kg, which indicates that 50% of the pumpkins are lighter than 2.7 kg and 50% of the pumpkins are heavier than 2.7 kg.
c 0.80 kg
d 0.5 kg ($1.35 \leq x \leq 4.15$)
e 2.77 kg
f 0.37 kg
g 0.23 kg
h median
i The standard deviation is 0.43 kg higher when the outlier is removed.
j interquartile range

3 a The average hourly wage earned by the group is $16.10.
b $7.60
c The middle hourly wage earned by the group is $14, which indicates that 50% of the group earns less $14 and 50% of the group earns more than $14.
d $39 ($2.5 \leq x \leq 26.5$)
e $14.20
f $3.30
g The median because it is not affected by the outlier
h The standard deviation decreases by 4.3 when the outlier is removed.

© Cengage Learning Australia Pty Ltd 2013